

Global Volatility Transmission: A Hierarchical Bayesian VAR Approach

1. Motivation and Economic Rationale

In highly integrated global financial markets, risk and uncertainty rarely stay confined to a single region. A volatility shock in the US market often quickly transmits to European and Asian markets. Modeling these transmission mechanisms is crucial for international portfolio risk management. While standard Vector Autoregressions (VARs) are the traditional tool for modeling such multivariate dynamics, they suffer from the "curse of dimensionality." As we add more global indices to the model, the number of parameters grows quadratically, leading to severe overfitting. This project proposes using a Bayesian VAR to solve this exact over-parameterization problem, allowing us to robustly model the joint dynamics of global volatility.

2. Econometric Methodology

We will implement the Bayesian Vector Autoregression methodology utilizing the hierarchical prior selection framework from Kuschnig and Vashold (2021).

Shrinkage via the Minnesota Prior: Directly applying the concepts from Week 3 lecture on "Shrinkage priors in regression analysis," we will use the Minnesota prior to shrink the dense parameter space of the VAR toward a parsimonious baseline (such as a random walk).

Hierarchical Structure: Instead of manually and subjectively setting the hyperparameters that dictate the "tightness" of this shrinkage, we will use the BVAR package's MCMC capabilities to treat these hyperparameters as random variables. The model will automatically learn the optimal degree of shrinkage from the data.

3. Data Application (April 16 Deliverable)

We will construct a multivariate system of global equity market volatility to showcase the primary analytical tools provided by the BVAR package.

Data Source: Daily implied volatility indices (e.g., VIX for the US, V2TX for Europe, VNKY for Japan, VFTSE for the UK) easily retrievable via yfinance or Bloomberg.

Analysis & Visual Output: The core of the April 16 presentation will focus on standard macroeconomic outputs, demonstrating how the BVAR package seamlessly computes them:

- **Impulse Response Functions (IRFs):** We will shock the US VIX by one standard deviation and plot the dynamic response of the European and Japanese volatility indices over a 30-day horizon. This visually demonstrates the speed and magnitude of volatility contagion.
- **Forecast Error Variance Decomposition (FEVD):** We will decompose the forecast error variance of each market. For instance, we will present a stacked bar chart showing what percentage of the European V2TX's variance at a 10-day horizon is driven by domestic European shocks versus imported shocks from the US.

- **Sub-sample Comparison:** To make the economic narrative engaging, we will estimate the model on two separate sub-samples (e.g., the calm period of 2017-2019 versus the volatile COVID-19 period of 2020-2022) to see how the IRFs and FEVDs structurally change during market crises.

4. Relevance to the Bayesian Econometrics Syllabus

This project directly addresses the core learning outcomes of the course:

- **Bayesian Regression & Shrinkage:** It serves as a practical, high-dimensional application of the shrinkage estimation discussed in Week 3.
- **Heteroskedasticity and Volatility:** It fulfills the requirement to "apply multivariate models for capturing heteroskedasticity in financial time series" by directly modeling variance indices.
- **Computational Approaches:** It demonstrates a state-of-the-art computational approach to Bayesian econometrics, specifically the implementation of hierarchical priors in R.